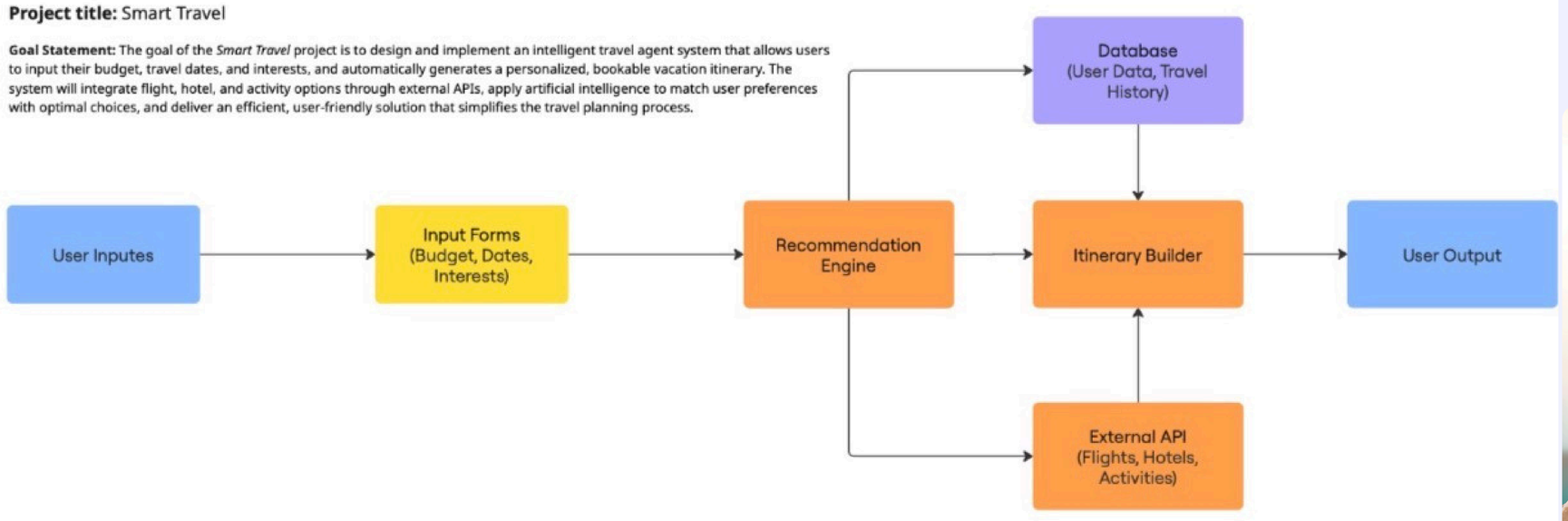
SmartTravel

SmartTravel simplifies trip planning using AI. Users enter travel preferences, and the system generates optimized itineries including flights, hotels, and activities. It uses NLP to interpret input, integrates with booking APIs, and applies AI for recomendations and budget optimization

DESIGN DIAGRAM

Project title: Smart Travel

Goal Statement: The goal of the *Smart Travel* project is to design and implement an intelligent travel agent system that allows users to input their budget, travel dates, and interests, and automatically generates a personalized, bookable vacation itinerary. The system will integrate flight, hotel, and activity options through external APIs, apply artificial intelligence to match user preferences with optimal choices, and deliver an efficient, user-friendly solution that simplifies the travel planning process.



FEATURES

User Experience

Personalized travel recommendations

Based on saved preferences (budget, interests, travel style)

Favorites system

For quick access to preferred destinations and repeat planning

Customizable itineraries

With ability to add notes and manage trip stages (draft → confirmed → completed)

Streamlined search experience

Across flights, hotels, and activities in one place

Challenges Addressed

API rate limiting required optimization strategies for high-traffic scenarios.

Data synchronization across devices presented challenges resolved through cloud architecture.

Core Functionality

User Management System

Enables secure user registration and login while maintaining personalized profiles with saved preferences.

Search & Discovery

Allows users to search flights, hotels, and activities based on location, dates, and interests.

Filtering System

Provides advanced filters such as budget, travel style, and interests to refine search results.

AI Recommendation Engine

Uses a multi-factor scoring algorithm to rank flights, hotels, and activities based on value, preferences, and relevance.

Itinerary Management

Users create, update, and manage itineraries by adding flights, hotels, and activities with status tracking.

System Integration

Integrated multi-service search

(flights, hotels, activities in one platform)

Centralized recommendation engine

Combining multiple data sources

User preference integration

Into search and recommendation pipeline

Real-time scoring and ranking system

Across all travel components

Future Development

Expand to include train and bus booking integration for comprehensive multi-modal travel planning.

Implement AR features for destination previews and navigation assistance.

TECHNOLOGY STACK

Frontend

React 18
Vite
Tailwind CSS
React Router
Axios

Backend

Python 3.9+
FastAPI
SQLAlchemy
Pydantic
SQLite / PostgreSQL

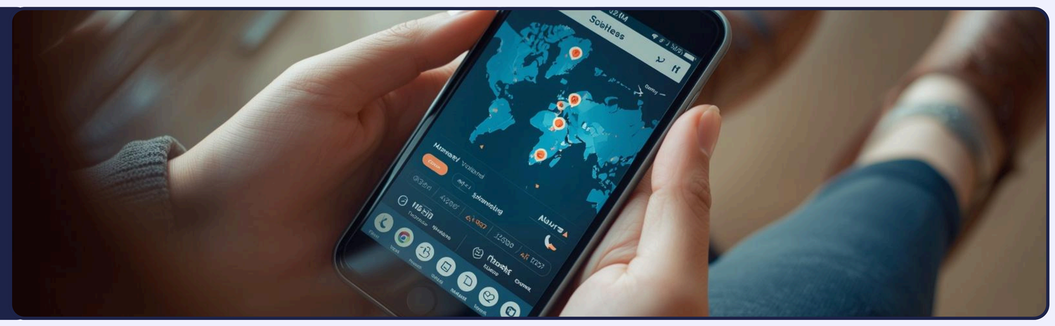
APIs Integrated

Amandeus API (flights & hotels)
Viator API (activities)
Google Places API (points of Interest)

"Travel smarter, not harder - where every journey is personalized, optimized, and effortlessly planned." ✈️

MEET THE TEAM

Sethu Kruthin Nagari - CS
Sai Venkata Subhash Vakkalagadda - CS

FACULTY ADVISOR

Dr. Nitin Nitin